

Pattern Program 1: Right-Angled Triangle of Stars

◆ Problem Statement:

Write a C program to print a right-angled triangle of * stars for a given number of rows.

◆ Sample Output (for 5 rows):

```
*
**
***
****
*****
```

◆ C Program:

```
#include <stdio.h>
int main() {
    int i, j, rows;
    printf("Enter number of rows: ");
    scanf("%d", &rows);

    for(i = 1; i <= rows; i++) {
        for(j = 1; j <= i; j++) {
            printf("*");
        }
        printf("\n");
    }
    return 0;
}
```

◆ Pattern Program 2: Number Pyramid

◆ Problem Statement:

Write a C program to print a number pyramid pattern based on the number of rows.

◆ Sample Output (for 5 rows):

```
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
```

◆ C Program:

```
#include <stdio.h>

int main() {
    int i, j, space, rows;
    printf("Enter number of rows: ");
    scanf("%d", &rows);

    for(i = 1; i <= rows; i++) {
        for(space = 1; space <= rows - i; space++) {
            printf(" ");
        }
        for(j = 1; j <= i; j++) {
            printf("%d ", j);
        }
        printf("\n");
    }
}
```

```

        }
        for(j = 1; j <= i; j++) {
            printf("%d ", j);
        }
        printf("\n");
    }

    return 0;
}

```

1. Hollow Square Pattern

Sample Output:

```

*****
*   *
*   *
*   *
*   *
*****

```

C Program:

```

#include <stdio.h>

int main() {
    int i, j;
    for(i = 1; i <= 5; i++) {
        for(j = 1; j <= 5; j++) {
            if(i == 1 || i == 5 || j == 1 || j == 5)
                printf("*");
            else
                printf(" ");
        }
        printf("\n");
    }
    return 0;
}

```

2. Number Right-Angled Triangle

Sample Output:

```

1
1 2
1 2 3
1 2 3 4
1 2 3 4 5

```

C Program:

```

#include <stdio.h>

int main() {
    int i, j;
    for(i = 1; i <= 5; i++) {
        for(j = 1; j <= i; j++) {

```

```
        printf("%d ", j);
    }
    printf("\n");
}
return 0;
}
```

3. Alphabet Triangle Pattern

Sample Output:

```
A
A B
A B C
A B C D
A B C D E
```

C Program:

```
#include <stdio.h>

int main() {
    int i, j;
    char ch;
    for(i = 1; i <= 5; i++) {
        ch = 'A';
        for(j = 1; j <= i; j++) {
            printf("%c ", ch++);
        }
        printf("\n");
    }
    return 0;
}
```

4. Diamond Star Pattern

Sample Output:

```
  *
 ***
*****
 ***
  *
```

C Program:

```
#include <stdio.h>

int main() {
    int i, j, space;
    int n = 3;

    for(i = 1; i <= n; i++) {
        for(space = 1; space <= n - i; space++)
            printf(" ");
        for(j = 1; j <= 2 * i - 1; j++)
```

```

        printf("*");
        printf("\n");
    }
    for(i = n - 1; i >= 1; i--) {
        for(space = 1; space <= n - i; space++)
            printf(" ");
        for(j = 1; j <= 2 * i - 1; j++)
            printf("*");
        printf("\n");
    }
    return 0;
}

```

5. Floyd's Triangle

Sample Output:

```

1
2 3
4 5 6
7 8 9 10
11 12 13 14 15

```

C Program:

```

#include <stdio.h>

int main() {
    int i, j, num = 1;
    for(i = 1; i <= 5; i++) {
        for(j = 1; j <= i; j++) {
            printf("%d ", num++);
        }
        printf("\n");
    }
    return 0;
}

```

1. Right-Aligned Star Triangle

Sample Output:

```

      *
     **
    ***
   ****
  *****

```

C Program:

```

#include <stdio.h>

int main() {
    int i, j, space;
    for(i = 1; i <= 5; i++) {

```

```

        for(space = 1; space <= 5 - i; space++)
            printf(" ");
        for(j = 1; j <= i; j++)
            printf("*");
        printf("\n");
    }
    return 0;
}

```

2. Inverted Number Triangle

Sample Output:

```

1 2 3 4 5
1 2 3 4
1 2 3
1 2
1

```

C Program:

```

#include <stdio.h>

int main() {
    int i, j;
    for(i = 5; i >= 1; i--) {
        for(j = 1; j <= i; j++) {
            printf("%d ", j);
        }
        printf("\n");
    }
    return 0;
}

```

3. Binary Number Triangle

Sample Output:

```

1
0 1
1 0 1
0 1 0 1
1 0 1 0 1

```

C Program:

```

#include <stdio.h>

int main() {
    int i, j;
    for(i = 1; i <= 5; i++) {
        for(j = 1; j <= i; j++) {
            printf("%d ", (i + j) % 2);
        }
        printf("\n");
    }
}

```

```
    return 0;
}
```

4. Half Diamond Star Pattern

Sample Output:

```
*
**
***
**
*
```

C Program:

```
#include <stdio.h>

int main() {
    int i, j;
    for(i = 1; i <= 3; i++) {
        for(j = 1; j <= i; j++)
            printf("*");
        printf("\n");
    }
    for(i = 2; i >= 1; i--) {
        for(j = 1; j <= i; j++)
            printf("*");
        printf("\n");
    }
    return 0;
}
```

5. Hollow Right-Angled Triangle

Sample Output:

```
*
**
* *
*  *
*****
```

C Program:

```
#include <stdio.h>

int main() {
    int i, j;
    for(i = 1; i <= 5; i++) {
        for(j = 1; j <= i; j++) {
            if(j == 1 || j == i || i == 5)
                printf("*");
            else
                printf(" ");
        }
        printf("\n");
    }
}
```

```
    }  
    return 0;  
}
```

1. Reverse Pyramid of Stars

Sample Output:

```
*****  
*****  
*****  
***  
*
```

C Program:

```
#include <stdio.h>  
  
int main() {  
    int i, j, space;  
    int rows = 5;  
    for(i = rows; i >= 1; i--) {  
        for(space = 0; space < rows - i; space++)  
            printf(" ");  
        for(j = 1; j <= 2*i - 1; j++)  
            printf("*");  
        printf("\n");  
    }  
    return 0;  
}
```

2. Continuous Characters Triangle

Sample Output:

```
A  
B C  
D E F  
G H I J  
K L M N O
```

C Program:

```
#include <stdio.h>  
  
int main() {  
    char ch = 'A';  
    int i, j;  
    for(i = 1; i <= 5; i++) {  
        for(j = 1; j <= i; j++) {  
            printf("%c ", ch++);  
        }  
        printf("\n");  
    }  
    return 0;  
}
```

```
}
```

3. Right-Angled Triangle with Even Numbers

Sample Output:

```
2
2 4
2 4 6
2 4 6 8
2 4 6 8 10
```

C Program:

```
#include <stdio.h>

int main() {
    int i, j;
    for(i = 1; i <= 5; i++) {
        for(j = 1; j <= i; j++) {
            printf("%d ", 2 * j);
        }
        printf("\n");
    }
    return 0;
}
```

4. Triangle of Odd Numbers

Sample Output:

```
1
3 5
7 9 11
13 15 17 19
21 23 25 27 29
```

C Program:

```
#include <stdio.h>

int main() {
    int i, j, num = 1;
    for(i = 1; i <= 5; i++) {
        for(j = 1; j <= i; j++) {
            printf("%d ", num);
            num += 2;
        }
        printf("\n");
    }
    return 0;
}
```

5. Hollow Pyramid Pattern

Sample Output:

```
  *
 * *
*   *
*     *
*****
```

C Program:

```
#include <stdio.h>

int main() {
    int i, j, space;
    int rows = 5;
    for(i = 1; i <= rows; i++) {
        for(space = 1; space <= rows - i; space++)
            printf(" ");
        for(j = 1; j <= 2*i - 1; j++) {
            if(j == 1 || j == 2*i - 1 || i == rows)
                printf("*");
            else
                printf(" ");
        }
        printf("\n");
    }
    return 0;
}
```